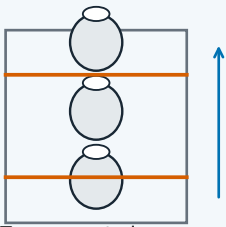
Global observation and local verification

Fixed detection budget for embedded cattle passage counting

Observe



Two separated zones

Allocate

Global image

Next global image
or same-image crop

Same static detector graph
Equal real call quota

Count and time

PYNQ-Z2

Directional passage totals
Actual global / local calls
Publication age and queue

Wall-paced source replay

Temporary tracking IDs support counting; persistent cow identity is a separate task.